

Yash K. Jawale

POSTDOC · TU DELFT

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Research Interests

Cytoskeleton, Biophysics, Quantitative Biology, Synthetic Biology, Microscopy, Image analysis, Computation

Skills

Biochemical & Molecular Techniques	<i>in vitro</i> reconstitutions, encapsulation, protein purification, gene cloning
Experimental & Imaging Techniques	Microscopy (confocal, TIRF, IRM), microfluidics, optogenetics
Programming & Computational Tools	Python, MATLAB, Cytosim, ImageJ/Fiji, R, GROMACS
Other Technical Skills	LaTeX, FreeCAD (3D design), 3D printing, Blender (SciViz)

Publications

7. **Jawale, Y.K.**, Schröeter, M.[‡], De Lora, J.[‡], Shourabi, A. Y.[‡], De Franceschi, N., Aubin-Tam, M.-E., Spatz, J. P., & Dogterom, M. (n.d.). *Tubulin acts as a surfactant and disrupts lipid membranes* [in preparation]
6. Roth, S., Gârlea, I. C., Vleugel, M., **Jawale, Y. K.**, Mulder, B. M., & Dogterom, M. (n.d.). *Reconstitution of basic mitotic spindles in cell-like confinement* [under review]
5. Van de Cauter, L.[‡], **Jawale, Y. K.**[‡], Tam, D., Baldauf, L., van Buren, L., Koenderink, G. H., Dogterom, M., & Ganzinger, K. A. (2024). High-speed imaging of giant unilamellar vesicle formation in cDICE. *ACS Omega*, 9(41), 42278–42288. [10.1021/acsomega.4c04825](https://doi.org/10.1021/acsomega.4c04825)
4. Chaphalkar, A. R.[‡], **Jawale, Y. K.**[‡], Khatri, D., & Athale, C. A. (2021). Quantifying intracellular particle flows by DIC Object Tracking. *Biophysical Journal*, 120(3), 393–401. [10.1016/j.bpj.2020.12.013](https://doi.org/10.1016/j.bpj.2020.12.013)
3. Joshi, Y. J., **Jawale, Y. K.**, & Athale, C. A. (2020). Modeling the tunability of the dual-feedback genetic oscillator. *Physical Review E*, 101(1). [10.1103/physreve.101.012417](https://doi.org/10.1103/physreve.101.012417)
2. **Jawale, Y. K.**, Rapol, U., & Athale, C. A. (2018). Open Source 3D printed focussing mechanism for cell phone based cellular microscopy. *Journal of Microscopy*, 273(2), 105–114. [10.1111/jmi.12765](https://doi.org/10.1111/jmi.12765)
1. Beal, J., Haddock-Angelli, T., Gershater, M., de Mora, K., Lizarazo, M., Hollenhorst, J., & Rettberg, R. (2016). Reproducibility of Fluorescent Expression from Engineered Biological Constructs in E. coli. *PLOS ONE*, 11(3), e0150182. [10.1371/journal.pone.0150182](https://doi.org/10.1371/journal.pone.0150182)

Thesis

2. **Jawale, Y.K.** (2025, September). *Building Minimal Spindles: Reconstituting spindle positioning in synthetic cells* [Ph.D. dissertation]. Delft University of Technology. [10.4233/UUID:F4E25106-CB82-4706-90B7-88800D45C8D2](https://doi.org/10.4233/UUID:F4E25106-CB82-4706-90B7-88800D45C8D2)
1. **Jawale, Y. K.** (2018, May). *Role of external forcing in directional instability of microtubule transport* [M.Sc. thesis]. IISER Pune. <http://dr.iiserpune.ac.in:8080/xmlui/handle/123456789/987>

[‡]Equal contributions

Research Experience

Doctoral work Asymmetric force generation in Synthetic cell Marileen Dogterom group	TU Delft, NL	Jan. 2020 - Present
Project Assistant Open Source Image Analysis of Sub-Cellular Microscopy Data: Leveraging Big Data Astronomy Tools	IISER Pune, IN	Aug. 2018 - Dec. 2019
Master's work Role of External Forcing in Directional Instability of Microtubule Transport Chaitanya Athale group (GPA 9.0 (of 10))	IISER Pune, IN	Jun. 2017 - Apr. 2018
Semester Project(s) <i>In vitro</i> Reconstitution and Quantification of recombinant Kinesin (KHC) Gliding Assay	IISER Pune, IN	Aug. 2016 - Apr. 2017
Semester Project(s) Optimizing the Foldscope: with PDMS and Glass lens microscopes, and Focusing Mechanism	IISER Pune, IN	Aug. 2015 - Apr. 2016
Summer Project iGEM (International Genetically Engineered Machine) Mycobacterium Revelio (Wiki) (Bronze Medal)	IISER Pune, IN	May 2015 - Sept. 2015

Education

PhD Delft University of Technology (TUD)

Delft, NL September 2025

BS-MS (Dual degree) Indian Institute of Science Education and Research (IISER) Pune (CGPA 8.3 (of 10))

Pune, IN May 2018

HSC Maharashtra State Board (84.33%)

Ahmednagar, IN February 2013

SSC Maharashtra State Board (87.27%)

Ahmednagar, IN March 2011

Conferences

EMBO Microtubules: From Atoms to Complex Systems (POSTER)

EMBL Heidelberg, DE Jun. 2026

EMBO Frontiers in Cytoskeleton Research (POSTER)

IISER Pune, IN Nov. 2025

EMBO Microtubules: From Atoms to Complex Systems (POSTER)

EMBL Heidelberg, DE Jun. 2024

SynCell International Conference on Engineering Synthetic Cells and Organelles

Toulouse, FR Apr. 2024

ASCB | EMBO Cell Biology (POSTER)

Boston, USA Dec. 2023

NWO Dutch BioPhysics (POSTER)

Veldhoven, NL Oct. 2023

DivIDE 2 Spindle Structure: From Protein to Architecture (TALK & POSTER)

CRG Barcelona, ES Sept. 2023

Naito 49 Frontiers of microtubule and its-related motors: atomic structures, cellular functions, development and diseases (TALK & POSTER)

Sapporo, JP Jul. 2023

SynCell International Conference on Engineering Synthetic Cells and Organelles (TALK)

Minneapolis, USA May 2023

HRZZ Mitotic spindle: From living and synthetic systems to theory (POSTER)

Dubrovnik, HR Apr. 2023

NWO Dutch BioPhysics (POSTER)

Veldhoven, NL Oct. 2022

EMBO Microtubules: From Atoms to Complex Systems (POSTER)

EMBL Heidelberg, DE Jun. 2022

SynCell International Conference on Engineering Synthetic Cells and Organelles (POSTER)

The Hague, NL May 2022

EMBO Microtubules: From Atoms to Complex Systems (POSTER | VIRTUAL)

EMBL Heidelberg, DE Jun. 2020

EMBO Creating is Understanding: Synthetic Biology Masters Complexity (FLASH TALK & POSTER)

EMBL Heidelberg, DE Sept. 2019

Indian Biophysical Society Emerging Trends in Biophysics (POSTER)

IISER Pune, IN Mar. 2018

M2T2 Current Trends in Transport and Molecular Motor (POSTER)

TIFR, Mumbai, IN Dec. 2017

Natural and Artificial Molecular Machines (POSTER)

IIT-Bombay, Mumbai, IN Dec. 2017

Biophysical Society USA Computational and Experimental Studies of Microtubules and Microtubule based Motor Proteins (POSTER)

IIT-Bombay, Mumbai, IN Dec. 2016

iGEM Gaint Jamboree, International Genetically Engineered Machine (TALK & POSTER | BRONZE MEDAL)

Boston, USA Sept. 2015

Workshops

Lorentz Center Reconstituting Biology: Charting the Way to Minimal Cells
Microscopy Image Analysis

Lieden, NL Jul. 2022

Biophysical Society USA Modern Biophysical Tools and Techniques
Dynamics of Complex Systems (Mathematical Biology)

NCCS, Pune, IN Mar. 2019

IIT-Bombay, Mumbai, IN Dec. 2017

ICTS, Bangalore, IN May 2017

Teaching

Judge International Genetically Engineered Machine (iGEM)
Assessing and grading the iGEM student teams at the Gaint Jamboree.

Paris, FR Oct. 2022

Teaching Assistant Physical Biology of the Cell II
Designing assignments and conducting work-lectures.

TU Delft, NL 2021-2022

Teaching Assistant Research Practical | Optical Tweezers
Conducting practical and analysis sessions.

TU Delft, NL 2020-2021

Teaching Assistant BioPhysics
Designing assignments, exams and solutions.

TU Delft, NL 2020

Session Coordinator B4 Workshop on Bioimaging: Advanced Light Microscopy
Conducted a MATLAB based image analysis session.

IISER Pune, IN May 2019

Guiding the team with experimental design, modeling, analysis and presentation. Won Silver Medal.

Achievements

INSPIRE Fellowship (MHRD, GOVT. OF INDIA)

IN 2013-2018

GATE Life Sciences (AIR (ALL INDIA RANK): 188)

IN 2018

CSIR-UGC Life Sciences (LECTURESHIP (RANK: 35))

IN 2017

Miscellaneous

Member Association for Socially Applicable Research ([ASAR](#))  ([Related works](#))

IN Jun. 2018 - Present

Pro bono work - Data curation, analysis and visualizations.

European Synthetic Cell Initiative Highlight | Collaboration: Yash and Jacqueline ([blog](#))

NL Feb. 2023

iGEM Community Alumni Features | A Pursuit to Figuring Out How Biology Works: It Began at iGEM ([blog](#))

IN Feb. 2023

Invited Talk UGC HRDC Synthetic Biology: Refresher Course (Virtual)

IN Jan. 2021

Hobbies

• Numismatics • Philately • Homelabbing & self-hosting • Cycling